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Problem Chosen
Wildlife Protection
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IM²C
Summary Sheet

Planning Wildlife Protection under Geographic and Resource Constraints

Protecting a large reserve requires a plan that links important locations, completed inspections and timely ground response. We study how Etosha National Park can allocate personnel and drones, maintain service through seasonal tasks, and adapt the framework to other parks.

Method. We divide Etosha into 17 reporting regions and 599 planning units. Historical animal surveys and burnable-habitat areas define relative priorities, while road routes and a two-hour response rule determine where inspections can receive effective follow-up. A linear allocation model maximizes weighted task completion. A monthly extension adds fire checks and maintenance of 17 historically documented boreholes, then finds the personnel needed to maintain the selected service target. The Effective Inspection Completion Rate measures prescribed inspections completed with timely ground response, weighted by conservation priorities; it does not measure wildlife survival.

Allocation results. The baseline assumes that 60% of 295 reference employees are available: 177 person-months, or **21240** effective personnel-hours. Thirty proposed drones supply **1200** flight-hours per month. The maximum effective inspection completion rate is **57.26%**; a representative solution uses **5379.02** personnel-hours, including response readiness, and **151.82** flight-hours. The remaining capacity shows that access and response locations limit service. Among three tested additional posts, the ENP5 candidate raises the rate to **64.27%** within the same total budgets.

Seasonal demand and robustness. Maintaining the target requires **57** personnel in the busiest normal month and **62** under the specified drought-maintenance scenario. These are conditional workload estimates. Tested animal-habitat weights change the priority-weighted percentage but leave the base deployment unchanged; tighter budgets make priorities matter more. Travel speed, response access and visit timing also affect service.

Recommendation and transfer. Prioritize verified response access and workable visit dates before adding equipment. Trials for Chitwan and Yellowstone use real spatial inputs and explicit local assumptions; Chitwan's boundary fails the area-consistency check, so its results remain conditional polygon trials. Actual staffing and ecological adequacy require field validation.

Keywords: wildlife protection; resource allocation; ground response; seasonal workforce; Etosha National Park.

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Reading guide. Sections 4–9 answer Questions 1–6. The letter uses nontechnical language. References and the AI-use report are included within this document’s 25-page ceiling.

1. Introduction

Large wildlife reserves face a practical protection problem: important locations are spread across a wide area, while staff, equipment and travel time are limited. An observation becomes useful only when an appropriate team can follow it up. More patrol hours or more drones may increase monitoring activity without making distant locations easier to reach. The problem therefore asks how conservation priorities should be defined, how ground teams and drones should be allocated, and how the workforce should change as seasonal tasks increase. Etosha National Park provides the starting case, with a reported area of about 22,935 km² and a reference workforce of 295 employees [1].

We address the six questions through one connected planning framework. First, we define a measurable inspection service and identify the conservation concerns supported by the available evidence. We then allocate ground and drone work subject to travel, response and resource limits. A monthly extension estimates the personnel needed to maintain the same basic service while completing additional fire checks and borehole maintenance. Sensitivity tests examine reduced resources, altered travel conditions and different visit dates. Finally, we assess the model's limitations and test how its structure can be adapted to parks in Asia and North America. The purpose is to support transparent deployment decisions with a clear account of what the results mean.

Our spatial method organizes Etosha into 17 reporting regions and 599 inspection units. Historical animal counts, legal protection categories and burnable-habitat areas establish relative task priorities; mapped roads and candidate response posts determine travel time. Each unit has a representative location and a prescribed monthly inspection demand. A task receives effective service only if its screening is completed and a qualified ground team can arrive within two hours. We combine these completion levels using normalized importance weights to obtain the Effective Inspection Completion Rate, expressed from 0 to 100%. It measures completion of prescribed inspections with timely ground response; wildlife survival and ecological losses require separate evidence.

A linear program chooses the ground personnel-hours, drone flight-hours and completion level for each unit. Its constraints account for inspection capacity, drone suitability, operating personnel, shared response readiness and a minimum service requirement for the reachable part of every region. The baseline assumes that 60% of the reference workforce is available for these tasks at 120 effective hours per person-month, providing 21240 personnel-hours. Thirty proposed drones provide 1200 flight-hours per month. These are planning conditions rather than verified operating rules. By treating personnel and flight time as separate resources, the model also makes explicit that drone observations still require people for transport, operation and follow-up.

The seasonal model retains the basic inspection target and adds month-specific fire screening and maintenance visits to 17 historically documented boreholes. It first calculates the extra workload, then minimizes total personnel-hours needed to meet the target under the shared flight budget. Dividing that workload by 120 and rounding upward gives a monthly personnel requirement; the largest month determines a fixed available team size. The drought case changes maintenance frequency and on-site time, rather than forecasting rainfall, animal populations or water-delivery volumes. We separately check whether pooled monthly work can conceal long gaps between visits.

The baseline effective inspection completion rate is 57.26%. A representative allocation uses 5379.02 personnel-hours and 151.82 flight-hours, leaving substantial resources unused because the existing response layout limits credited service. Of three additional-post candidates, ENP5 raises the completion rate to 64.27% under unchanged total resource limits. Maintaining the baseline target requires 57 available personnel in the normal peak month and 62 in the specified drought-maintenance case. Both estimates are conditional on the tasks included. Trials for Chitwan and Yellowstone retain the task, route and response structure while rebuilding local inputs; Chitwan's boundary-area mismatch prevents treating its result as calibrated park staffing. The main conclusion is that personnel, equipment and access should be planned together, with field validation preceding operational recommendations.

2. Basic Assumptions and Justifications

The following assumptions define a planning scenario rather than verified park operating rules.

Assumption 1: Work is pooled monthly, using 120 effective hours per person, 60% personnel availability and 30 drones.

Justification: Declared limits permit comparable deployment tests; monthly feasibility does not prove a daily roster.

Assumption 2: Inspection demand and within-region responsibility are allocated by area, with equal fire exposure on burnable habitat.

Justification: Fine animal locations and multiyear harmful-fire exposure are unavailable; explicit proxies avoid inventing observations.

Assumption 3: Ground and drone teams complete the same checklist only where suitable; credited service needs ground arrival within two hours and shared teams handle infrequent alerts.

Justification: Observations require follow-up. Task equivalence, service standards and concurrency still need field calibration.

3. Data Description and Symbol Definitions

The problem gives an approximate area of 22,935 km²; our projected boundary covers 22897.10 km². We use 2015 animal surveys, 2021 land cover and a fixed 2026 road snapshot [1, 2, 4, 5], organized into 17 reporting regions and 599 inspection units. A representative location supports route calculations, not continuous area coverage. Question 2 excludes water and rainfall; Question 3 separately adds fire screening and maintenance of 17 historical boreholes without forecasting populations or water-delivery volumes.

A personnel-hour is one person's work for one hour; a flight-hour is one aircraft's flight for one hour. The Effective Inspection Completion Rate measures prescribed inspections completed with timely ground response, weighted by conservation priorities. It is separate from ecological outcomes and area coverage.

Table 1: Main symbols and their meanings.

Symbol	Definition
i, j, t	Reporting region, inspection unit and month.
A_j, C_j	Unit area and prescribed monthly checks.
w_j, s_j	Relative inspection importance and completion fraction.
P, η	Effective inspection completion rate and target, expressed in percent.
r_j	1 when ground staff can arrive within the time limit; 0 otherwise.
x_j, h_j	Ground screening personnel-hours and drone flight-hours.
H, U, H_0	Personnel limit, flight limit and reserved response work.

4. Protection Priorities and a Measurable Standard

4.1 Priorities and evidence

Our provisional priorities use evidence of threats, potential loss, affected extent and the possibility of intervention. They are management judgments rather than a measured ranking of incident probabilities.

Table 2: Provisional management priorities.

Priority	Challenge and evidence
1	Poaching and illegal entry: the problem identifies continuing rhinoceros losses; priority-object losses are difficult to replace.
2	Harmful wildfire and habitat damage: potentially widespread effects on animals and habitats; distinguish prescribed fire.

Human–wildlife conflict, disease, important plants and birds remain data needs. Missing information does not make them low-priority. Black rhinoceros remains a key conservation concern even though the current regional animal-priority measure lacks reliable spatial counts for that species. The six measured survey objects are blue wildebeest, plains zebra, giraffe, African savanna elephant, gemsbok and ostrich.

Historical abundance is first expressed as a regional share within each species. Legal protection categories then provide a basis for relative responsibility. This avoids treating a large number of common animals as automatically more important than a smaller priority population. Burnable grassland, shrubland and forest areas provide a separate habitat responsibility measure. Only a single fire-data sample was quality checked, so the base model assumes equal exposure per unit of burnable habitat rather than estimating a multiyear harmful-fire risk.

4.2 What effective inspection completion measures

A target receives effective service when prescribed screening is completed and a qualified ground team can arrive within the selected response limit. Let $d_j \in [0, 1]$ describe screening completion and $r_j \in \{0, 1\}$ describe response eligibility. For normalized demand weights w_j , define

$$\begin{aligned}
 g_j^{\text{eff}} &= d_j r_j, \\
 P &= 100 \sum_j w_j g_j^{\text{eff}}, \\
 L &= \sum_j w_j (1 - g_j^{\text{eff}}).
 \end{aligned} \tag{4.1}$$

The allocation model represents effective completion directly by $s_j \leq r_j$, giving the same Effective Inspection Completion Rate P . Here P is reported in percent: $P = 57.26$ means 57.26%. With fixed weights, increasing P reduces unmet weighted demand L . This priority-weighted rate differs from an area completion rate and does not measure the proportion of animals protected.

Actual protection needs independent evidence: illegal-loss records with comparable monitoring effort, population surveys, habitat condition and the impact of harmful fires. More detected incidents may reflect better detection. Prescribed burning is not automatically protection failure. A park average cannot certify missing species or replace their dedicated safeguards.

5. Resource Allocation with Ground Teams and Drones

5.1 Planning units and resource assumptions

The 15 survey regions, main salt pan and other survey-external areas form 17 reporting regions $\mathcal{I}$. A 10 km grid intersected with each region produces 599 connected inspection units $\mathcal{J}$. Each has an interior representative location and support area A_j . The location is a route-planning proxy, not proof of continuous coverage of that area.

Six mapped locations serve as candidate response posts: Okaukuejo, Halali, Namutoni, Olifantsrus, Galton Gate and Nehale Gate. Their coordinates are sourced; their use as staffed response posts is assumed. We model monthly personnel availability, not a limit requiring a particular fraction of employees to be on duty simultaneously.

Table 3: Etosha operating assumptions and resource settings.

Item	Setting
Travel speed	Road 30, walking 4, drone flight 40 km/h.
Task time / team	Observation 10, preparation 5, dispatch 10 minutes; two people per team.
Service / response	8 checks per 100 km ² /month; ground arrival within 2 hours.
Technology eligibility	At most 0.6 flight-hours/check; regional openness at least 60%.
Access / service floor	Entry-index decay scale 2 hours; reachable-area service floor 15%.
Personnel	295 reference employees; 60% availability; 120 hours/person-month.
Drones	30 aircraft; 2 flight-hours/day for 20 flyable days/month.
Readiness	Six candidate posts, two people each, 8 hours/day for 30 days.

The two resource limits and shared response reservation are

$$\begin{aligned}
 H &= 295(0.60)(120) = 21240 \text{ personnel-hours/month,} \\
 U &= 30(2)(20) = 1200 \text{ flight-hours/month,} \\
 H_0 &= 6(2)(8)(30) = 2880 \text{ personnel-hours/month.}
 \end{aligned} \tag{5.1}$$

The available inspection labor before task allocation is $H - H_0 = 18360$ hours. Response readiness is charged once for the park. Staffing fractions, flight allowances and service standards are explicit planning assumptions; they are not verified operating rules at Etosha.

5.2 Animal responsibility, habitat and access

For species m , let N_{im} be its surveyed abundance in region i , $q_{\ell(m)}$ its category weight, and $\mathcal{I}_S$ the 15 surveyed regions. Let A_i^F denote burnable-habitat area. Define

$$\begin{aligned}
 V_i^A &= \frac{\sum_m q_{\ell(m)} (N_{im} / \sum_{v \in \mathcal{I}_S} N_{vm})}{\sum_m q_{\ell(m)}}, \\
 V_i^F &= \frac{A_i^F}{\sum_v A_v^F}.
 \end{aligned} \tag{5.2}$$

Hunttable, protected and specially protected categories use the team comparison matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 1/2 \\ 3 & 2 & 1 \end{pmatrix}, \tag{5.3}$$

$$\mathbf{q} = (0.163424, 0.296961, 0.539615)^\top.$$

The normalized principal eigenvector gives $\mathbf{q}$; $CR = 0.007933 < 0.1$ checks consistency, not ecological truth. The law supplies categories, not these numerical comparisons [3].

For unit j , the shortest entry time from one of 27 road–boundary intersections is $T_j^E = D_j^E/v_g + d_j^\perp/v_w$. Its access index is $E_j = \exp(-T_j^E/2)$ with time in hours. These are potential access sources, not proven poacher entrances. Within a region, $\mu_j = A_j/A_i$ apportions responsibility by area: $R_j^A = V_i^A \mu_j E_j$ and $R_j^F = V_i^F \mu_j$.

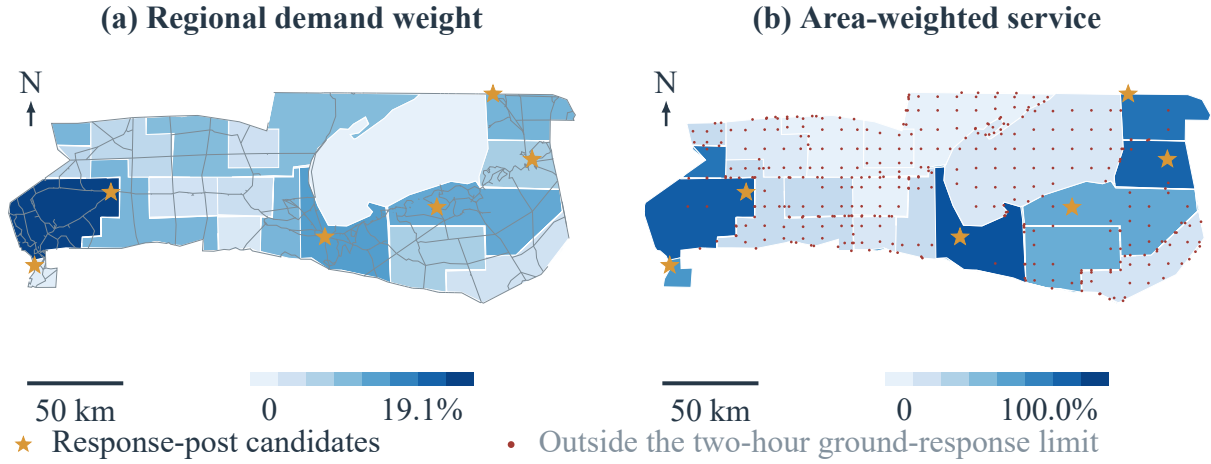


Figure 1: Etosha demand and base area service; stars are candidate posts, red dots fail the response rule.

Animal responsibility is unknown outside the survey. Zero coefficients there mean exclusion from the measured-animal component, not zero animal value. The salt pan's zero fuel baseline also does not imply zero ecological value.

5.3 Weights, routes and response eligibility

We estimate the two component weights over the common 15-region scope. First aggregate $X_{ik} = \sum_{j \in \mathcal{J}_i} R_j^k$, then calculate

$$\begin{aligned} p_{ik} &= \frac{X_{ik}}{\sum_v X_{vk}}, \\ e_k &= -\frac{\sum_i p_{ik} \ln p_{ik}}{\ln 15}, \\ \alpha_k &= \frac{1 - e_k}{(1 - e_A) + (1 - e_F)}. \end{aligned} \quad (5.4)$$

With $0 \ln 0 = 0$, $e_A = 0.881596$ and $e_F = 0.934989$ give $\alpha_A = 0.645551$ and $\alpha_F = 0.354449$. Entropy weights express spatial differences in the chosen data, not observed loss shares. Each component is separately normalized over its observed scope, then $w_j = \alpha_A w_j^A + \alpha_F w_j^F$ and $W_i = \sum_{j \in \mathcal{J}_i} w_j$. Thus $\sum_j w_j = 1$.

The directed road graph retains original connections and one-way restrictions. Outbound and return distances are calculated separately. A post qualifies only if its round trip is finite and

$$\begin{aligned} t_d + d_{bj}^+/v_g + d_j^\perp/v_w &\leq T_{\max}, \\ r_j &= \mathbf{1}\{\text{at least one qualifying post}\}. \end{aligned} \quad (5.5)$$

We choose the qualifying post with the shortest road round trip $D_j = d_{bj}^+ + d_{jb}^-$. Response eligibility concerns the outbound journey; task cost also includes the return. Authorized use of candidate roads and straight-line off-road travel are assumptions requiring field checks.

$$\begin{aligned}
a_j &= n_g(D_j/v_g + 2d_j^\perp/v_w + t_o + t_p), \\
b_j &= 2d_j^\perp/v_u + t_o, \\
o_j &= n_u(D_j/v_g + t_p + b_j), \\
\gamma_j &= o_j/b_j.
\end{aligned} \tag{5.6}$$

Ground cost a_j and drone-support cost o_j are personnel-hours per check; b_j is flight-hours per check. Drone use h_j therefore consumes $\gamma_j h_j$ personnel-hours. Only response-eligible units meeting endurance and openness conditions enter $\mathcal{J}_U$.

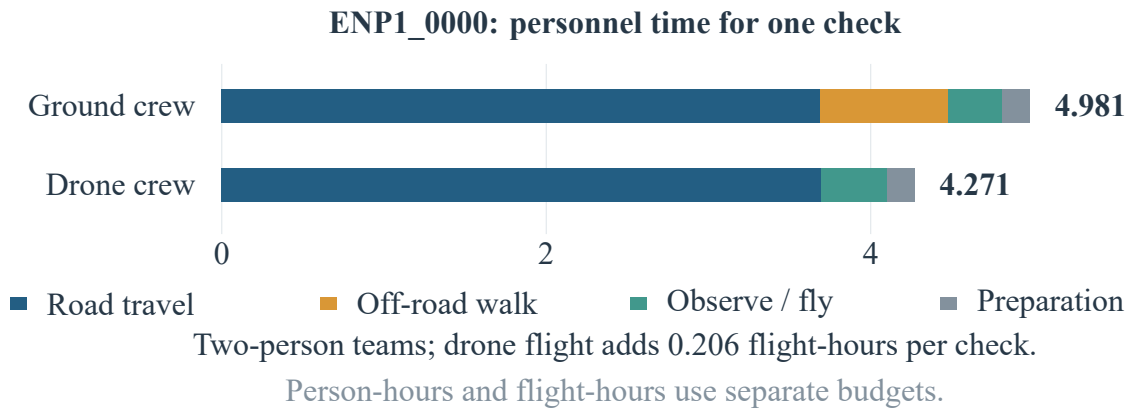


Figure 2: Personnel work per check at ENP1_0000; drone flight time is a separate resource.

5.4 Service capacity and the allocation model

Decision variables are ground screening hours x_j , drone flight-hours h_j , and effective completion s_j . The prescribed monthly checks are based on area:

$$\begin{aligned}
C_j &= 8A_j/100, \\
\sum_j C_j &= 8 \sum_j A_j/100.
\end{aligned} \tag{5.7}$$

Areas are in km^2 , giving 1831.77 prescribed checks per month. Refining the grid redistributes area without creating additional total demand. A completed check is a task at a representative location. Fractional checks describe planning intensity and require integer dispatches for implementation.

Define capacity coefficients $g_j = 1/a_j$ for eligible ground tasks, otherwise zero; $u_j = 1/b_j$ for eligible drone tasks, otherwise zero. Then $C_j s_j \leq g_j x_j + u_j h_j$. This avoids dividing by undefined costs and permits different checks to be shared between technologies without double-counting the same inspection.

For each region, $Q_i = \sum_{j \in \mathcal{J}_i} \mu_j r_j$ is reachable area share and $\bar{s}_i = \sum_{j \in \mathcal{J}_i} \mu_j s_j$ is area-weighted completion. A common floor $\bar{s}_i \geq 0.15 Q_i$ protects basic service within the reachable part. If only 20% of a region is reachable, the full-region floor is 3%, not 15%.

$$\begin{aligned}
& \max_{\mathbf{x}, \mathbf{h}, \mathbf{s}} \quad 100 \sum_j w_j s_j \\
& \text{s.t.} \quad C_j s_j \leq g_j x_j + u_j h_j, \quad j \in \mathcal{J}, \\
& \quad \sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j \leq H - H_0, \\
& \quad \sum_j h_j \leq U, \\
& \quad \sum_{j \in \mathcal{J}_i} \mu_j s_j \geq 0.15 Q_i, \quad i \in \mathcal{I}, \\
& \quad 0 \leq s_j \leq r_j, \quad j \in \mathcal{J}, \\
& \quad x_j \geq 0, \quad j \in \mathcal{J}, \\
& \quad h_j \geq 0, \quad j \in \mathcal{J}, \\
& \quad x_j = 0, \quad r_j = 0, \\
& \quad h_j = 0, \quad j \notin \mathcal{J}_U.
\end{aligned} \tag{5.8}$$

The fixed coefficients make this a linear program. Completion is capped at one so excess inspections cannot raise the completion rate above 100%. Human and flight budgets have different units and are linked through operator labor, not added together.

Increasing completion by Δs_j needs $C_j a_j \Delta s_j$ ground labor or $C_j o_j \Delta s_j$ drone-support labor plus $C_j b_j \Delta s_j$ flight time. Hence high demand alone does not determine allocation: workload, access and both budgets also matter.

5.5 Solution checks and regional allocation

HiGHS solves 1797 variables. Independent checks recalculate task capacity, response and flight eligibility, both budgets, region floors and totals. Mathematical feasibility does not validate road permissions, observation effectiveness or an actual roster.

After maximizing the effective inspection completion rate, we fix the returned service vector $\mathbf{s}^{(1)}$ and minimize $H_M = \sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j$. This selects the least labor supporting that particular service pattern. It does not search all equal-rate inspection patterns for the global minimum workforce.

The effective inspection completion rate is 57.26%. The selected representative uses 5379.02 personnel-hours and 151.82 flight-hours, including 2880 response hours and 2499.02 drone-support hours. Its ground screening hours are zero; ground operators and response staff are still essential.

Demand percentages are $100W_i$; reach and service percentages are $100Q_i$ and $100\bar{s}_i$. Regional labor excludes shared response readiness. ENP1 receives 487.89 screening-support hours. ENP11 has 8.05% demand weight but only about 0.02% reachable area. Additional ordinary inspection resources alone cannot remove this gap.

The salt pan's unknown animal value is not filled with zero. Its current objective weight is zero, while its generic service floor still applies. Equal-rate inspection choices there can change representative labor consumption without changing the park completion rate.

Figure 3 locates the resource totals for all 17 regions in Table 4. The six candidate posts each reserve 480 response personnel-hours per month. Regional shading includes monitoring and operator support; it excludes those shared readiness hours. ENP9 receives no effective service because no unit meets the response rule. ENP11 has substantial demand but almost no reachable area. The map reports hours, not integer employees or aircraft stationed in each region. Post locations are fixed assumptions rather than an optimized station network.

Table 4: Regional demand and the representative base allocation.

Region	Demand (%)	Reach (%)	Service (%)	Labor (h)	Flight (h)
ENP1	19.06	87.87	87.87	487.89	29.31
ENP12	11.02	92.49	92.49	385.64	26.87
ENP15	9.96	52.33	52.33	352.52	19.90
ENP17	8.84	77.60	77.60	214.11	15.58
ENP3410	8.71	19.21	19.21	155.98	8.38
ENP11	8.05	0.02	0.02	0.26	0.01
ENP16	5.93	84.63	84.63	183.34	14.00
ENP13	5.88	49.87	49.87	282.75	13.08
ENP2	4.50	76.71	76.71	92.86	4.44
ENP5	4.39	5.94	5.94	30.79	0.80
ENP7	3.31	0.19	0.19	0.52	0.02
ENP9	2.90	0.00	0.00	0.00	0.00
ENP14	2.63	11.67	11.67	72.40	2.06
ENP6	2.55	20.57	20.57	60.50	3.62
ENP8	1.54	8.55	8.55	14.93	0.75
Other*	0.72	61.81	61.81	62.52	4.19
Salt pan*	0.00	9.05	8.80	102.01	8.82

* Animal values unknown. Regional labor excludes shared readiness. Small positive values below reporting precision are marked < 0.01 .

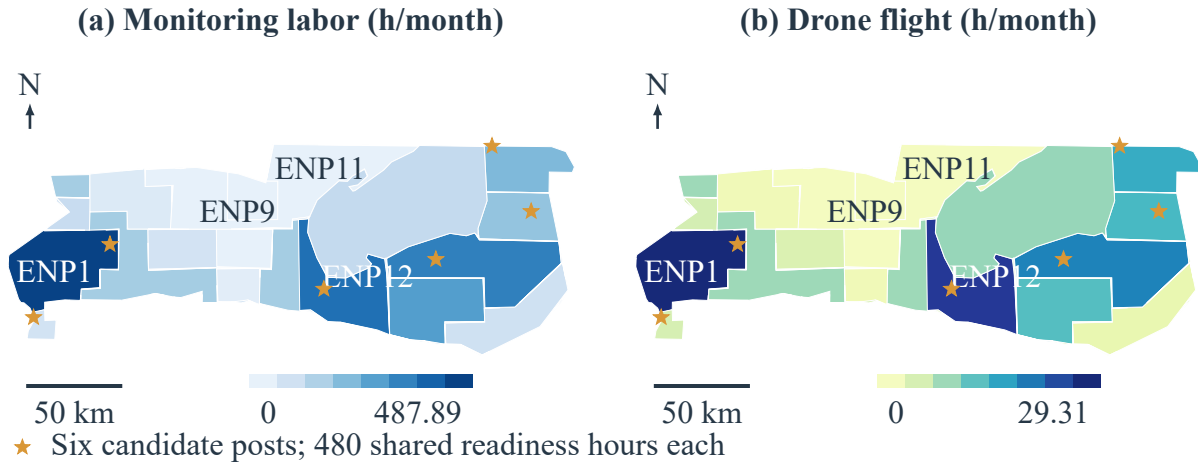


Figure 3: Optimal resource deployment under fixed candidate posts: monthly monitoring labor and drone flight-hours. Shared response readiness is separate.

5.6 Strategic comparisons and response posts

Since $w_j \geq 0$ and $s_j \leq r_j$, the geographic upper bound on effective inspection completion is

$$P \leq P_{\text{geo}} = 100 \sum_j w_j r_j. \quad (5.9)$$

The base reaches this bound with 15860.98 personnel-hours and 1048.18 flight-hours unused. Of 599 units, 216 meet the response standard, covering 34.12% of projected area. The 57.26% effective inspection completion rate uses priority weights; the 34.12% figure measures response-eligible area.

We compare the optimized allocation with uniform completion over reachable units and completion proportional to regional demand density W_i/A_i . Costs, resource limits and floors are identical.

All three methods reach the bound even after a 40% personnel-budget reduction. Thus this ample-resource comparison provides no completion-rate advantage for optimization. Removing drones still achieves 57.26%, but the returned ground-only plan uses 6244.08 personnel-hours. Under equal team

Table 5: Effective inspection completion rates (%) under equal budgets and service rules.

Budget	Labor limit	Uniform	Demand-based	Optimized
Base	21240	57.26	57.26	57.26
Personnel budget -40%	12744	57.26	57.26	57.26

sizes and checklist times, drones replace slower off-road walking; this saves labor only where the task is technically suitable.

Each tested additional post adds 480 readiness hours within the same total budgets. Routes and response eligibility are recalculated, with demand weights fixed.

Table 6: Additional-post candidates within the base total budgets.

Layout	Readiness (h)	Rate (%)	Total labor (h)	Flight (h)
Original six posts	2880	57.26	5379.02	151.82
ENP11 added	3360	58.86	5981.94	161.54
ENP3410 added	3360	58.95	5918.01	158.51
ENP5 added	3360	64.27	6294.98	189.25

The ENP5 candidate improves the completion rate by 7.01 percentage points. It is best among three tested locations, not a global siting optimum. Verify access, permissions and operating conditions before treating the site as a recommendation.

6. Seasonal Service and Workforce Requirements

6.1 Target, months and additional tasks

Question 3 asks how service changes through time and how many people are needed to maintain a chosen level. We use the same weights, inspection standard, routes, response locations and technical eligibility as Question 2. The target is its achieved optimum of 57.26%, using the unrounded percentage value $\eta = 57.26016128$; its ecological sufficiency is not established by being mathematically optimal.

Each planning month has 30 days and 120 effective personnel-hours per person. Reference availability is 177 people and 1200 monthly flight-hours. Assignments may change between months. Existing equipment, operators and response staff cannot be charged twice for concurrent tasks.

The dry-season classification is May–October in the cited study; the official fire strategy distinguishes May–July and August–November operating windows [9, 6]. These sources support the timing, not the numerical task frequencies. All frequencies below are team service rules.

Table 7: Medium service rules by month.

Months	Water visits/point	Extra fire checks*
Jan–Apr, Dec	2	0
May–Jul	4	2
Aug–Oct	4	4
Nov	2	4

*Checks per 100 km² of burnable habitat per month; on-site maintenance is 0.5 hour/visit in normal months.

Extra fire checks are distinct tasks added to the base checklist. Let A_j^{fuel} be the unit's support area multiplied by its region's observed burnable-habitat fraction. Then

$$D_{jt} = k_t \frac{A_j^{\text{fuel}}}{100} r_j. \quad (6.1)$$

The same inspection protocol and cost coefficients apply. These are screening tasks, not firefighting or prescribed burning. The factor r_j restricts credited tasks to the existing response range; excluded demand is separately recorded.

At the late-fire rate, 275.76 additional checks are supported inside that range, while 428.45 fall outside it. Adding personnel to the same six-post layout cannot guarantee those excluded tasks. We retain this gap rather than claiming park-wide fire coverage.

6.2 Borehole maintenance as ground-only work

A published 2013 field table contains 42 water-point records; 17 are classified as boreholes [7]. We use those 17 mapped points as a historical maintenance sample. It is not a complete current operational inventory. Natural springs are not assigned pumping-equipment work. Ministry records support solar-powered borehole equipment and maintenance as relevant activities [8].

Each visit includes a directed road round trip from a candidate post, any required walking, preparation and on-site inspection. We assume two people travel together. Let T_ℓ^{rt} be elapsed round-trip travel time, f_t visits per point and τ_t on-site time. Monthly personnel work is

$$\begin{aligned} a_{\ell t}^W &= 2(T_\ell^{\text{rt}} + \tau_t + 1/12), \\ W_t &= f_t \sum_{\ell \in \mathcal{L}} a_{\ell t}^W. \end{aligned} \quad (6.2)$$

The 1/12 hour term is five minutes of preparation. Every route is independently returned; no travel saving is claimed from combining visits. Drone observation does not replace maintenance of physical equipment.

Table 8: Historical-sample maintenance scenarios.

Rule	Visits/point	On-site hours	Monthly labor
Wet regular	2	0.5	211.17
Dry regular	4	0.5	422.34
Drought pressure	8	1.0	980.68

For the 17 points, summed elapsed round-trip travel is 42.87565 hours. The regular dry-month workload is

$$W_{\text{dry}} = 4(2) [42.87565 + 17(0.5 + 1/12)] = 422.34 \text{ personnel-hours.} \quad (6.3)$$

Wet-month work is 211.17 hours; the specified drought-maintenance rule produces 980.68 hours. These increases are visit and task assumptions, not inferred water shortages or observed fault rates.

No water-hauling volumes, groundwater response or animal-population changes are modeled. Major repairs, emergency pumping, unlisted points and shared-route opportunities need an updated inventory and operational records. The longest modeled regular visit takes about 6.39 elapsed hours, but fitting an eight-hour day for one task does not prove a complete monthly roster.

6.3 Shared monthly constraints

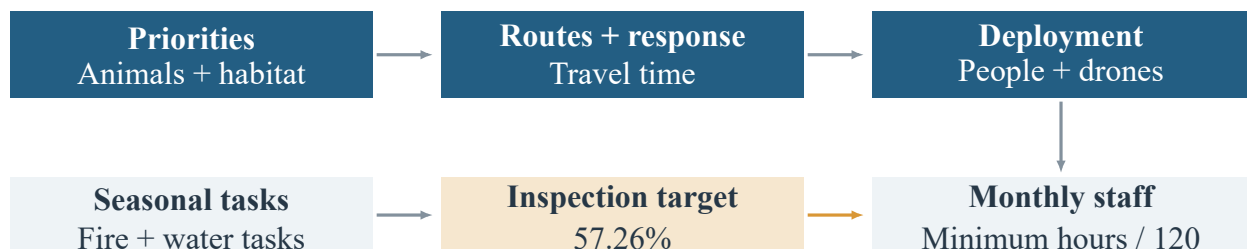
For each unit and month, (x_{jt}, h_{jt}, s_{jt}) serve the base inspection protocol. Extra fire work uses ground hours y_{jt} and flight-hours v_{jt} . These are separate task quantities with shared personnel and flight budgets. Where a technology is unavailable, its capacity term and investment are zero.

$$\begin{aligned}
 C_j s_{jt} &\leq g_j x_{jt} + u_j h_{jt}, \\
 D_{jt} &\leq g_j y_{jt} + u_j v_{jt}, \\
 \sum_j (h_{jt} + v_{jt}) &\leq U, \\
 H_t &= H_0 + W_t + \sum_j (x_{jt} + y_{jt}) + \sum_{j \in \mathcal{J}_U} \gamma_j (h_{jt} + v_{jt}).
 \end{aligned} \quad (6.4)$$

The remaining conditions retain $0 \leq s_{jt} \leq r_j$, nonnegative investments, technology eligibility and the regional floor

$$\sum_{j \in \mathcal{J}_i} \mu_j s_{jt} \geq 0.15 \sum_{j \in \mathcal{J}_i} \mu_j r_j. \quad (6.5)$$

Only base inspections enter $P_t = 100 \sum_j w_j s_{jt}$. Required fire and water tasks must also be completed; they do not change the base inspection completion rate's meaning. Readiness and maintenance are charged once, while operators serve both inspection types. Shared budgets prevent double allocation. Monthly totals support strategic planning; daily schedules, crew availability and simultaneous alerts still require checks. Thirty drones do not imply thirty simultaneous flights. Seasonal rules describe workload scenarios, not forecasts of animal abundance or drought.



Checks receive service credit only where a ground team can respond.

Figure 4: How priorities and task work connect to deployment and workforce.

6.4 Forward service and inverse workforce models

Two uses of the same feasible task set answer different questions. With N available people, the forward model finds the best monthly service. With a fixed service target, the inverse model finds the least total personnel work:

$$P_t^*(N, U) = \max_{\mathbf{z}_t \in \mathcal{F}_t(U), H_t \leq 120N} 100 \sum_j w_j s_{jt}, \quad (6.6)$$

$$H_t^{\min}(\eta, U) = \min_{\mathbf{z}_t \in \mathcal{F}_t(U), P_t \geq \eta} H_t.$$

Here $\mathbf{z}_t$ collects all five variable blocks and $\mathcal{F}_t(U)$ contains the shared constraints. The inverse model removes the reference labor ceiling so that a shortage can be expressed as a required workforce instead of only returning infeasibility. It allows different equal-rate inspection patterns, unlike the fixed-pattern refinement in Question 2.

$$N_t = \left\lceil \frac{H_t^{\min}}{120} \right\rceil, \quad (6.7)$$

$$N_{\text{year}} = \max_{t=1, \dots, 12} N_t.$$

The maximum, rather than the sum, estimates a constant workforce that meets the busiest month's pooled workload. It does not certify qualifications, exact simultaneous coverage or the completeness of the task inventory.

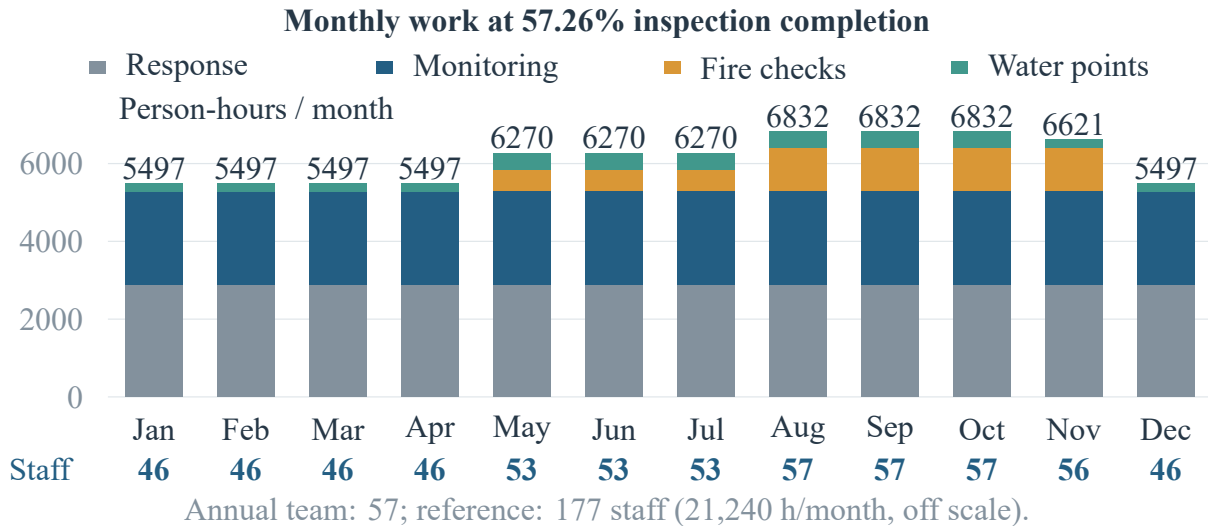


Figure 5: Medium-rule monthly workload and personnel equivalents; reference availability is 177 people.

With no added seasonal work and Question 2's returned service vector fixed, the calculation reproduces 5379.02 hours. Allowing the equal-rate pattern to change gives 5286.11 hours, a difference of 92.90. This comparison explains why dividing a single representative allocation by 120 need not yield the globally least personnel demand.

6.5 Normal-season results and feasibility

All months maintain the target with the 177-person reference and 1200 flight-hours. The least-work solutions need fewer personnel equivalents because only specified tasks and the existing response range are modeled.

The peak occurs in August–October. Its work separates into four parts:

$$H_{\text{peak}}^{\min} = 2880 + 2406.11 + 1123.38 + 422.34 = 6831.83 \text{ personnel-hours}. \quad (6.8)$$

These are readiness, base inspection labor, extra fire inspection labor and borehole maintenance. The flight requirement is 210.61 hours, well below 1200. Rounding $H_{\text{peak}}^{\min}/120$ gives 57 people. A check with 56 people returns 56.73, below the target, under the same monthly constraints.

Table 9: Minimum work and workforce for the medium service target.

Months	Personnel-hours	People	Flight-hours	Rate at 177 staff (%)
Jan–Apr, Dec	5497.28	46	144.34	57.26
May–Jul	6270.14	53	177.48	57.26
Aug–Oct	6831.83	57	210.61	57.26
Nov	6620.66	56	210.61	57.26

This is a minimum for the selected workload and pooled model. It is not evidence that the real park can reduce its workforce to 57 or that 177 people are currently assigned to protection. The 60% fraction is a resource scenario, and omitted threats, night service, specialist tasks and incident concurrency could change the required workforce.

Numeric checks independently recalculate each task's supported checks, prohibited allocations, both budgets, floors and aggregation. They confirm feasibility of the saved plans. Field calibration is a separate step: check actual task duration, travel permissions, maintenance status and whether drone and ground observations satisfy the same checklist.

The geographic completion-rate ceiling remains 57.26% throughout these months. Spare personnel capacity can meet extra tasks, but it cannot enlarge the fixed response range. A decision to raise the target above that ceiling first requires a change in access or response deployment.

6.6 Drought work and service-standard sensitivity

The abnormal-drought case increases maintenance visits during dry months from four to eight and on-site work from 0.5 to one hour. Timing is a defined workload scenario; it is not a forecast of drought, failures or rainfall. The peak becomes 7390.17 hours, or 62 personnel equivalents, with 210.61 flight-hours.

The extra work relative to the normal peak is 558.34 hours, mainly ground-only maintenance. The required workforce rises by 5 people relative to the 57-person normal plan. Both remain below the 177-person reference; no additional personnel gap is inferred from that reference.

Table 10: Workforce under different assumed service rules.

Rule	Normal people	Normal hours	Drought people	Drought hours
Low	51	6041.97	53	6270.14
Medium	57	6831.83	62	7390.17
High	71	8513.55	77	9207.89

The low, medium and high rules vary both visit requirements and extra fire checks. Their outcomes are planning alternatives, not confidence intervals. High service rules can use more flight-hours than the previous six-drone planning budget, while the current thirty-drone allowance supports these modeled tasks. A lower required headcount under a relaxed checklist must not be reported as unchanged ecological protection at lower cost.

Operational interpretation. Prepare workload and visit-date records before the busy months. Count response readiness and maintenance separately from screening. Verify whether a constant team can meet each month's work, then check daily task timing, training and specialist needs. Equipment inventory, available operator-hours and simultaneous flights are different quantities.

The historical 17-point sample is a material limitation. Replicating its workload tests sensitivity to inventory size, but does not establish additional point locations, routes or the actual number of functioning boreholes. Updating that inventory is more informative than adding unsupported water volumes to the equations.

7. Sensitivity and Scenario Analysis

Reducing baseline personnel-hours by 40% still gives 57.26% effective inspection completion. Lower resource budgets are tested with weights and tasks fixed. All 127 feasible saved solutions pass independent checks; four infeasible cases are excluded rather than plotted as zero completion.

Mid-level seasonal peak: joint personnel and flight capacity						
Staff	0	80	160	200	240	1200
45	20.3%	33.8%	37.2%	37.2%	37.2%	37.2%
55	44.7%	52.0%	55.2%	55.7%	55.7%	55.7%
57	47.7%	54.1%	57.0%	57.3%	57.3%	57.3%
62	53.6%	57.3%	57.3%	57.3%	57.3%	57.3%
67	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%
177	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%

Columns: flight-hours/month; cells: effective inspection completion (%).

Target 57.26%; 177 staff and 1200 flight-hours are the reference.

Figure 6: Peak-month effective inspection completion under joint personnel and flight limits; each cell is reoptimized.

At the medium peak, 57 people with 200 flight-hours maintain the target; 57 with 160 hours reach 56.97%. With no drones, the target needs 67 people. With 59 people and 120 flight-hours, reallocation reaches 57.24%, compared with 57.14% under the retained spatial service rule. Under the specified drought-maintenance pressure at the same low budgets, the gain from reallocation is 6.83 percentage points, but the target is still missed.

For the same 68 borehole visits and 422.34 monthly personnel-hours, staggered dates reduce the maximum gap from 27 to 8 days and the daily water-work peak from 105.58 to 15.48 hours. This water-only schedule is not a full roster or proof of a strict 7.5-day rule.

At road speeds of 20 and 40 km/h, the completion-rate ceilings are 35.76% and 64.69%. Extending response to three hours gives 80.64% under a changed service standard. These are conditional scenarios, not measured speed ranges or event probabilities.

7.1 Weight sensitivity

Reoptimization keeps other conditions fixed. Entropy measures spatial differences; CRITIC uses variation and indicator overlap. Table 11 distinguishes completion rates under each weight choice from evaluation using the original weights.

Table 11: Weight sensitivity: effective inspection completion (%) under each weight choice and original-weight evaluation.

Weight choice	Animal (%)	Own weights	Original weights	Tight budget
Entropy baseline	64.56	57.26	57.26	29.88
Equal components	50.00	53.18	57.26	29.87
Animal emphasis	90.00	64.40	57.26	29.88
CRITIC alternative	44.80	51.72	57.26	29.86

At 21240 personnel-hours, all four choices give identical deployment and a baseline completion rate of 57.26%. Reachable positive-weight demand is satisfied; response gaps remain.

The separate scarcity trial allows 3600 personnel-hours, including 2880 readiness hours, and 1200 flight-hours. Up to 9 units change completion relative to its entropy solution, while rates evaluated with the original weights span only 29.86–29.88%. Deployment responds to priorities, but the completion-rate effect is small in these tests.

Perturbing specially protected category preferences by $\pm 20\%$, with reciprocal comparisons and recalculated weights, preserves matrix consistency and base deployment. Tight-budget completion-rate losses

stay below 0.002 percentage points. All cases pass independent capacity, eligibility, regional-floor and budget checks; robustness applies to these tested choices.

8. Strengths, Limitations and Practical Use

8.1 Strengths

The framework connects explicit tasks to their travel, observation and operator work. It distinguishes a lack of resources from locations that fail the response standard. Ground and drone inputs share personnel capacity while retaining a separate flight limit. The inverse model explains seasonal workforce changes without forecasting animal populations.

All inputs and saved allocations are traceable. Independent checks cover capacity, floors, prohibited technology use, budgets, response gates and totals. A second calculation reproduces the cross-park optimization results. These checks establish computational consistency; they do not establish measured ecological outcomes.

8.2 Weaknesses and Limitations

The model depends on historical surveys, assumed task costs and unverified access conditions. Monthly pooled hours do not establish a feasible daily roster. Its effective inspection completion rate also cannot confirm sufficient ecological protection. Table 12 identifies the main limits and the observations needed to address them.

Table 12: Limits that affect interpretation and deployment.

Limit	Consequence and data needed
Historical / missing objects	Current priority-species locations and surveys are needed; six measured groups do not certify all wildlife.
Service proxies and assumptions	Calibrate check frequency, category priorities, fire exposure and ground/drone checklist equivalence.
Maps and permissions	Verify roads, barriers, seasons and emergency access; a mapped route is not confirmed permission.
Monthly pooled work	Check dates, qualifications, integer visits and concurrent incidents before calling a roster feasible.
Incomplete maintenance inventory	Update borehole status, tasks and repair duration; the 17-point sample cannot define all current work.
Service versus ecological outcomes	Validate loss, habitat and population outcomes independently; optimal service does not prove adequate protection.

First verify current priority-species and maintenance inventories. Then record actual travel, observation and response times, road permissions and equipment use. Compare ground and drone completion of the same checklist in suitable units. Check visit dates and simultaneous tasks after the monthly workload is feasible.

For implementation, preserve generic service floors and add dedicated safeguards for priority objects whose distribution is missing from the weighted completion measure. Use comparable ecological and incident records to test outcomes while controlling for observation effort. More data should revise task priorities and costs; it should not be used only to justify a previously chosen equipment count.

9. Adaptation to Parks on Other Continents

9.1 Shared structure and local data

We examine Chitwan National Park in Asia and Yellowstone National Park in North America. Priority setting, task capacity, response gates, readiness budgets and workforce conversion can remain. Species, habitats, seasonal routes, task protocols, permissions and operating costs must be rebuilt locally. Etosha’s animal weights and thirty-drone assumption are not transferred.

The spatial trial uses published boundaries, mapped roads and candidate locations, and WorldCover 2021. It assigns land-area weights $w_j = A_j^{\text{land}} / \sum_k A_k^{\text{land}}$ and $C_j = 8A_j^{\text{land}}/100$. Drones are set to zero because authorization and task equivalence are not calibrated. These choices test spatial service, not a complete evaluation of species protection.

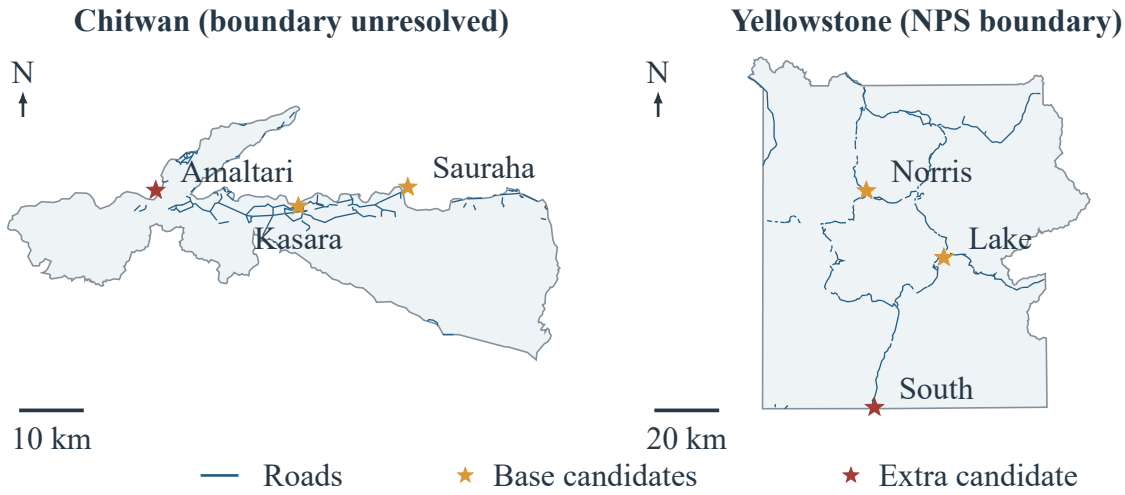


Figure 7: Published spatial inputs and candidate locations; Chitwan boundary consistency remains unresolved.

Table 13: Spatial trial inputs; Chitwan polygon remains uncalibrated.

Input	Chitwan polygon	Yellowstone NPS
Land service area (km ²)	1175.65	8470.07
Forest share (%)	91.11	62.93
Mapped park roads (km)	248.29	619.60
Units / grid side	1376 / 1 km	1520 / 2.5 km

Boundary restriction. The Chitwan polygon is 1214.06 km², versus 952.63 km² in the official source: a 27.44% difference. The published polygon is used only for a conditional trial; it is not accepted as the current legal park boundary. Yellowstone uses the NPS boundary. Both parks still need locally verified emergency access, off-road travel and task costs [12, 13, 11, 10, 14].

9.2 Conditional trials and expected deployment changes

For each park, fix its target at 90% of its own two-post, 30 km/h response-limited ceiling:

$$P_{\text{geo}}^{(p)} = 100 \sum_j w_j^{(p)} r_j^{(p)}, \quad (9.1)$$

$$\eta^{(p)} = 0.90 P_{\text{geo,base}}^{(p)}.$$

The target stays fixed when routes or speed change. It is not a 90-point whole-park target. Two-person teams, 120 hours/person-month, two-hour response and inspection times remain declared assumptions. A two-post readiness budget is 960 personnel-hours; a third post adds 480. Local incident response, community work and visitor management are not yet quantified in these personnel equivalents.

Table 14: Conditional land-inspection trials; ceiling and target are percentages, not actual park staffing.

Case	Setting	Ceiling	Target	Labor (h)	People
C0	Two posts, 30 km/h	37.57	33.81	1085.15	10
C1	20 km/h	30.66	33.81	Infeasible	–
C2	Third post added	45.58	33.81	1550.39	13
C3	40 km/h	42.12	33.81	1069.56	9
Y0	Two posts, 30 km/h	18.72	16.85	1541.63	13
Y1	20 km/h	12.02	16.85	Infeasible	–
Y2	Third post added	21.60	16.85	1973.23	17
Y3	40 km/h	24.48	16.85	1467.97	13

Both 20 km/h scenarios place the fixed target above the resulting ceiling, so extra labor alone cannot meet it. Additional posts improve eligible area but also increase readiness work. Higher speed is a sensitivity setting, not permission to drive faster than safe operating conditions allow.

In Chitwan, forest screening, priority-species work, river crossings and monsoon access should shape ground-team tasks and seasonal posts. In Yellowstone, visitor–wildlife interactions, winter routes and specialist transport should change the task list and feasible travel graph. Seasonal closure data for visitors must not automatically be treated as emergency-staff permissions. Drone operations require local authorization.

Halving unit size changes the two ceilings by about -0.13 and -0.20 percentage points. Finer land-cover sampling changes forest shares by about -0.002 and -0.056 percentage points. Ten numerical cases pass independent checks, while Chitwan’s boundary-consistency check still fails. These workforce equivalents are neither actual staffing estimates nor grounds for comparing the parks’ ecological condition.

Letter to IMMC

To the International Mission for Monitoring Conservation

Subject: A practical protection plan for Etosha and other large reserves

Dear IMMC Decision-Makers,

We recommend connecting monitoring to timely ground response. Our Etosha study shows that staff and drones can remain unused while important locations are too distant for effective follow-up.

Our approach divides the park into planning areas, uses historical animal surveys and burnable habitat to guide priorities, and calculates task and travel time. Routes and response posts determine which checks receive support. Drones can save walking time in suitable open locations, but people must still transport and operate them and follow up reported problems.

The planning baseline uses 60% of 295 reference employees, or 177 available people at 120 effective hours each, thirty drones and two-hour ground arrival. These assumptions do not describe verified current staffing or equipment.

The highest effective inspection completion rate under these conditions is 57.26%. A representative month uses about 5379 personnel-hours, including readiness, and 152 flight-hours. Both are well below the resource limits. This rate combines task completion and ground-response eligibility using conservation-priority weights. Wildlife losses, population health and habitat conditions need separate evaluation.

Among three additional-post candidates tested within the same budgets, ENP5 raises the completion rate to 64.27%. It is a direction for field assessment, not a confirmed global optimum. Check routes, permissions, site conditions and response needs before adoption.

Seasonal work also needs preparation. Additional fire checks and borehole maintenance raise the modeled normal peak requirement to 57 people. The specified drought-maintenance scenario raises it to 62. Both fit within the 177-person reference, but cover only the selected daytime tasks, response range and historical maintenance sample.

Begin with a limited field trial: record travel, inspection, drone use and follow-up time; compare checklist completion by ground and drone-assisted teams; update the maintenance inventory; and verify access. Use these records to revise routes, response locations and task assumptions.

Turn the monthly plan into a calendar. Stagger borehole visits to avoid long service gaps, check daily capacity and retain qualified response staff. Thirty drones do not imply thirty simultaneous flights; operator, readiness and maintenance work all need time.

Trials in Chitwan and Yellowstone show how the structure can be adapted using local spatial records and operating assumptions. Chitwan's boundary disagrees with the official area, so its trial cannot support a calibrated staffing recommendation. Yellowstone requires information on visitor interactions and winter access. Neither trial establishes actual park staffing.

At Etosha, records come from different years, and some priority species, including black rhinoceros, lack detailed distributions. The model assumes occasional alerts rather than several serious simultaneous incidents. Specialist and night work may add personnel needs; a high inspection completion rate cannot resolve these gaps.

We propose three connected actions: validate field assumptions; examine response locations that could close important access gaps; and revise monthly assignments using verified records. Retain basic service in less prominent areas and dedicated safeguards for priority species. Measure progress through completed checks, reliable follow-up and independently observed wildlife and habitat outcomes.

Personnel, equipment and access should be planned together. The framework provides a transparent starting point, with decisions revised as reliable evidence improves.

Yours faithfully,

The Modeling Team

Team 510105201001220151

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- [15] OpenAI. *Codex*, AI coding assistant used in the October 2026 work session; GPT-6 model family, exact deployed version unavailable. Uses and verification are described in the AI-use report. Online source.

Report on Use of AI

Tool and identification. OpenAI Codex, based on the GPT-6 model family, was used in this work session [15]. The exact deployed model version was not exposed in the session metadata. Python/SciPy and XeLaTeX performed numerical optimization and document compilation; those outputs were checked separately from generated prose.

Uses. The assistant helped explain existing formulas, distinguish sourced records from planning assumptions, revise resource parameters specified by the team, synchronize seasonal calculations, shorten chapter drafts, translate the integrated manuscript into English, draft the nontechnical letter and produce numbered figures and tables. Parallel assistants worked on bounded chapter, data-synchronization and visualization tasks.

Representative instructions. The team requested: (1) explain each stage and formula in the existing Question 2 document; (2) keep other parameters unchanged, make 60% of personnel available and use thirty drones, then shorten Question 2 to eight pages with simple terminology; (3) retrieve remote updates, revise Question 3 to six pages, revise Question 6 similarly, and integrate Questions 1–6, the template sections and a letter into an English paper of at most 24 pages. The team supplied the control number used here.

Verification. Reported Question 2–4 allocations were recomputed or checked against saved inputs and explicit constraints. The cross-park chapter uses the retrieved real-space results and retains its failed Chitwan boundary check. Scripts verify units, capacities, budgets, prohibitions, floors, totals, references, figure/table labels and final page count. Earlier rendered versions were visually reviewed. The final terminology revision was compiled without a further PDF or layout audit at the team's request. A computational check is not a validation of real ecological outcomes or a detailed roster.

Limits and responsibility. Generated wording and model choices require team review. The manuscript identifies historical data, scenario assumptions and uncalibrated inputs; it makes no claim that AI supplied current animal distributions, verified staffing inventories, ecological-loss effects or authorized access permissions. Sources are cited separately from the team's calculations. The team remains responsible for accuracy, originality and compliance with the competition's disclosure rules.

The uploaded template's obsolete year, example notation, sample identifiers and unfinished claims were replaced with the current problem, confirmed team number and verified results. The project retains source data, model inputs, allocation arrays, code, provenance and numerical/layout audit records for review.